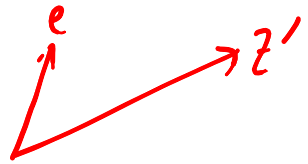
$$\frac{u^T z'}{\sqrt{n}} = \frac{1}{\sqrt{n}} \underbrace{\frac{u^T z'}{\|u\|}}_{\sim \mathcal{N}(0,1)} \cdot \overbrace{\|u\|}^{\sqrt{Q}}$$



$$e = \frac{u}{\|u\|} \qquad z' \sim \mathcal{N}(0, I)$$

$$w^T z \sim \mathcal{N}$$

$$\sim \mathcal{N}(w^T \mu, w^T \Sigma w)$$

$$\frac{1}{\sqrt{n}} \mathcal{N}(0,1) \sim \mathcal{N}(0, 1/n)$$

$$Q = \mu^T \Sigma^{-1} \mu = O(1) \quad \ni \quad \frac{1}{\sqrt{n}} \frac{u^T z'}{\|u\|} \sqrt{Q} \xrightarrow{p} 0$$

$$\frac{1}{n}\|z'\|^2 \qquad z' \sim \mathcal{N}(0, I)$$

$$\|z'\|^2 = z_1'^2 + \cdots + z_d'^2 \sim \chi_{\underset{\smile}{d}}^2$$

$$\begin{cases} \mathbb{E}_{X \sim \chi^2_d}(X) = d \\ \mathbb{V}(X) = 2d \end{cases}$$

$$\mathbb{E}\left(\frac{1}{n}\|z'\|^2\right) = d/n \qquad \mathbb{V}\left(\frac{1}{n}\|z'\|^2\right) = \frac{1}{n^2} 2d$$

$$= 2 \cdot d/n \cdot 1/n$$

$$\lim_{n\to\infty} d/n = d_0 \qquad = 2 \cdot d_0 \cdot 1/n \to 0$$

$$\frac{1}{n}\|z'\|^2 \xrightarrow{P} d/n \longrightarrow \alpha_0$$

$$\mathbb{P}(\text{Error} \mid D) \xrightarrow{P} \Phi\left(-\frac{Q}{\sqrt{Q+\alpha_0}}\right)$$

If $\alpha_0 = 0$ $\quad \mathbb{P}(\text{Error} \mid D) \xrightarrow{P} \underbrace{\Phi(-\sqrt{Q})}_{\text{Baye's Opt. Error}}$

$d = \text{Const.}$

$n \to \infty$

$d \neq \text{Const.}$ $\quad d/n \to \alpha_0 \neq 0$

$n \to \infty$

$\alpha_0 = 2$

$d = 100$ ... $d = 1000$ ... $d = 10^6$

$n = 50$ $\quad$ $n = 500$ $\quad$ $n = 5 \times 10^5$

$$\mu = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i Y_i$$

Ch. 2

Markov Ineq.

$X \geq 0$

$\forall t > 0 \qquad \mathbb{P}(X \geq t) \leq \dfrac{\mathbb{E}(X)}{t}$

Proof:

$$\mathbb{E}(X) = \int_0^\infty x \cdot f_X(x)\,dx \geq \int_t^\infty x \cdot f_X(x)\,dx$$

$$\geq \int_t^\infty t\, f_X(x)\,dx = t \cdot \int_t^\infty f_X(x)\,dx$$

$$= t \cdot \mathbb{P}(X \geq t)$$

$$\mathbb{P}\Big(\underbrace{(X-\mu)^2}_{Y \geq 0} \geq t^2\Big) \leq \frac{\mathbb{E}\{(X-\mu)^2\}}{t^2}$$

$$= \frac{\mathbb{V}(X)}{t^2}$$

$$\mathbb{P}(|X-\mu| \geq t) \leq \frac{\mathbb{V}(X)}{t^2}$$

$$e^{\lambda(X-\mu)} \qquad e^{\lambda t} \qquad \lambda > 0$$

$$\mathbb{P}\Big(\underbrace{e^{\lambda(X-\mu)}}_{Y \geq 0} \geq e^{\lambda t}\Big) \leq \frac{\mathbb{E}\{e^{\lambda(X-\mu)}\}}{e^{\lambda t}}$$

$$\mathbb{P}(X-\mu \geq t) \leq \frac{\mathbb{E}\{e^{\lambda(X-\mu)}\}}{e^{\lambda t}}$$

Moment Gen. Func.

$$\leq \inf_{\lambda > 0} \frac{\mathbb{E}\{e^{\lambda(X-\mu)}\}}{e^{\lambda t}}$$

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\mathbb{E}\left(e^{\lambda(X-\mu)}\right) = \int_{-\infty}^{+\infty} e^{\lambda(x-\mu)} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} \underbrace{e^{\lambda(x-\mu) - (x-\mu)^2/2\sigma^2}}_{\nearrow\, e^{-\frac{(x-a)^2}{2b^2} + c}} dx$$

$$b^2 = \sigma^2$$

$$a = \mu + \lambda\sigma^2$$

$$-\frac{(x-a)}{b^2} = \overbrace{\lambda - (x-\mu)/\sigma^2}^{-(x-\mu-\lambda\sigma^2)/\sigma^2}$$

$$e^{\lambda(x-\mu) - \frac{(x-\mu)^2}{2\sigma^2}}$$

$$x = a = \mu + \lambda\sigma^2$$

$$d\!\!\not{y} = \lambda^2\sigma^2 - \frac{\lambda^2\sigma^{\cancel{4}\,2}}{2\cancel{\sigma^2}} = \frac{1}{2}\lambda^2\sigma^2$$

$$c = \frac{1}{2}\lambda^2\sigma^2$$

$$M = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{c}\, e^{-(x-a)^2/2\sigma^2}\, dx$$

$$= e^{c} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-(x-a)^2/2\sigma^2}\, dx$$

$$M = e^{\lambda^2\sigma^2/2}$$

$$\mathbb{P}(X-\mu \geq t) \leq \frac{M}{e^{\lambda t}} = e^{\underbrace{\lambda^2\sigma^2/2 - \lambda t}}$$

$$\frac{\partial}{\partial\lambda}\left\{\lambda^2\sigma^2/2 - \lambda t\right\} = \lambda\sigma^2 - t = 0 \Rightarrow \lambda = t/\sigma^2$$

$$e^{\frac{t^2}{\sigma^4}\frac{\cancel{\sigma^2}}{2} - t^2/\sigma^2}$$

$$= e^{-t^2/2\sigma^2}$$

$$\mathbb{E}\left(e^{\lambda(X-\mu)}\right) \leq e^{\lambda^2\sigma^2/2}, \quad \lambda \in \mathbb{R}$$

X is Sub Gaussian

$$\mathbb{P}(X-\mu \geq t) \leq e^{-t^2/2\sigma^2}$$

$$\mathbb{P}(|X-\mu| \geq t) \leq 2 \cdot e^{-t^2/2\sigma^2}$$